

PAVAN KUMAR KOLA

Backend & AI Engineer | Python, FastAPI, Django, LLM Systems | Automation & Scalable API Architecture
+91 9618080752 | pavankumarkola752@gmail.com | linkedin.com/in/pavan-kumar-kola-991185278 |
github.com/Pawankumar-01

TECHNICAL SKILLS

Backend: Python (FastAPI, Django, Flask), ASP.NET Core Web API, REST API Design, Microservices Architecture
AI / ML: LLM Integration, RAG (Retrieval Augmented Generation), LangChain, NLP, Whisper STT, AI Workflow Automation
Frontend & Mobile: React.js, Next.js, Flutter
Databases: PostgreSQL, MongoDB, MySQL, Redis (Caching & Task Queues)
DevOps & Cloud: Docker, Celery, Azure, Google Cloud, Render, WSL Linux
Tools & Integrations: ERPNext, WhatsApp Business API, Instagram Graph API, JWT Auth, Git, Entity Framework
Languages: Python, JavaScript, C#, SQL

PROFESSIONAL EXPERIENCE

Software Engineer — Sai Ganga Panakeia Ltd

2024 - Present

Andhra Pradesh, India

- Architected and owned end-to-end development of scalable FastAPI backend services powering AI-driven clinical and enterprise workflows, with async processing pipelines supporting high-concurrency production loads.
- Designed and implemented RAG (Retrieval Augmented Generation) pipelines using LangChain to enable context-aware LLM reasoning over large internal knowledge bases, reducing response latency and improving answer relevance.
- Built modular LLM integration layers supporting multiple model backends, enabling intelligent summarization, decision support, and automated case-sheet generation from unstructured clinical conversations.
- Engineered PostgreSQL schemas and query-optimized data models supporting high-throughput transactional workloads across clinical ERP and patient management systems.
- Led design of event-driven microservice architecture using Redis and Celery for distributed background processing, enabling asynchronous AI pipeline execution at scale.
- Delivered full-stack integrations between Flutter mobile clients, ERPNext ERP infrastructure, and AI backend services — eliminating manual data fragmentation across clinical workflows.

PROJECTS

AI Project Management & Automation Platform

2024

Technologies: FastAPI, LangChain, Whisper, RAG, LLMs, PostgreSQL

- Architected a FastAPI backend as the central AI orchestration hub, designing async service layers to coordinate speech transcription, knowledge retrieval, and LLM inference in a unified pipeline.
- Developed production-grade RAG pipelines using LangChain to retrieve contextual knowledge from structured and unstructured data sources, enabling accurate, context-grounded LLM decision support.
- Integrated OpenAI Whisper for real-time speech-to-text transcription, building preprocessing pipelines to feed clean text into downstream NLP and summarization models.
- Designed modular, pluggable AI workflow architecture enabling rapid swapping of model backends and retrieval strategies without service disruption.

Integrated Clinical ERP Automation Platform

2024

Technologies: ERPNext, FastAPI, Flutter, PostgreSQL, REST APIs

- Designed and extended ERPNext healthcare modules to support custom clinical consultation workflows, governance-driven patient records, and automated data entry, eliminating fragmented manual processes.
- Built a high-performance FastAPI clinical engine as middleware between mobile consultation tools and ERP backend, handling business logic, validation, and bidirectional data synchronization.
- Engineered Flutter mobile interface with session-based consultation workflows enabling real-time data capture, AI transcription, and doctor-side fact validation in a single unified UX.
- Designed an event-driven integration layer propagating clinical state changes automatically to downstream ERP records, eliminating manual data duplication across departments.

EDUCATION

B.Tech in Computer Science Engineering

May 2020 – March 2023

Aditya College of Engineering and Technology, Andhra Pradesh, India

Diploma in Computer Science Engineering

June 2017 – March 2020

Andhra Polytechnic College, Andhra Pradesh, India

CERTIFICATIONS

- Django Application Development with SQL and Databases — August 2024
- Developing AI Applications with Python and Flask — July 2024
- Responsive Web Design — FreeCodeCamp, June 2023
- SQL — HackerRank, March 2024